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CMPM 125

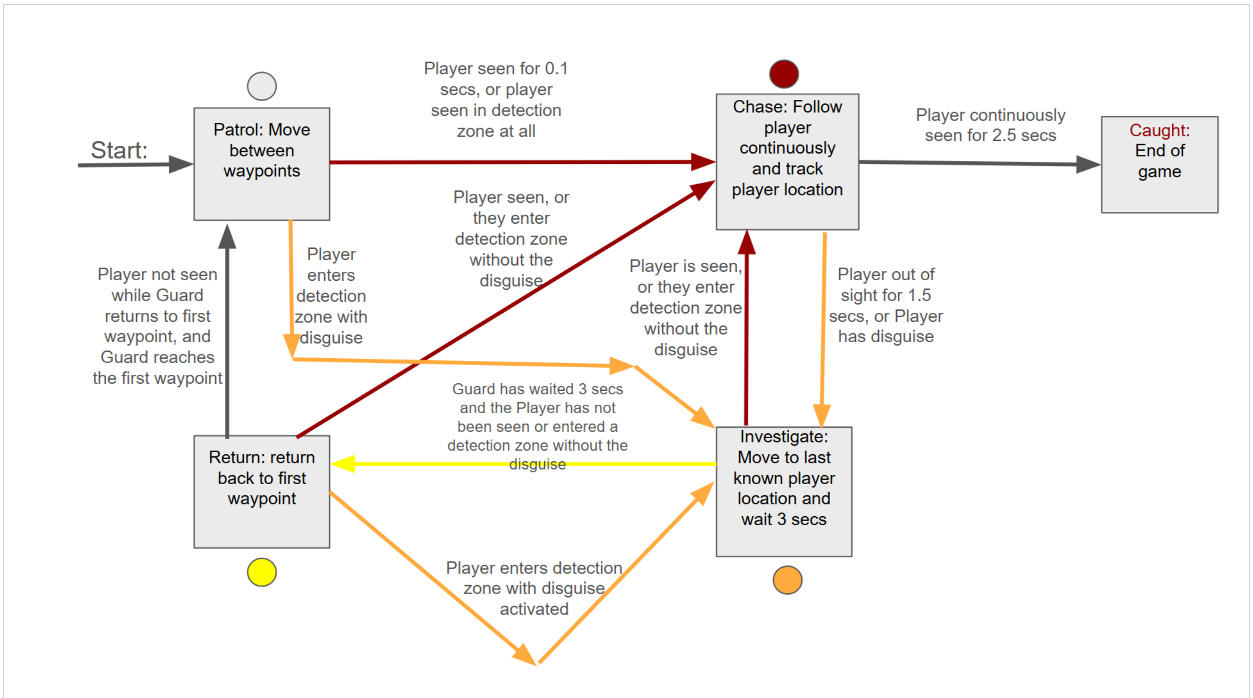
Markus Eger

Assignment 4: Report

AI implementation:

For our AI implementation our Guard AI uses a FSM with an enum and switch statement that runs every frame in update. It switches to a function for each state. The four states are Patrol, Chase, Investigate, and return with each having different transition logic and color as well as movement speed. The movement of the guards is done with navmesh. The Guard follows placed checkpoints and goes in a loop. For the detection of the player the Guards have a vision cone check with raycast to ensure obstacles block sightlines. There is also a PlayerInZone flag that handles the detection of when the player enters the light blue detection zone around the Guards. Finally there is a disguise mechanic where once it has been picked up the Guards will no longer chase you or see you. If you enter the detection zone with a disguise it will make the Guard investigate.

State Machine diagram:



Dominic: I made a simple map, created the navmesh and added mechanics such as a blue disguise item to prevent enemies from aggroing you and an item you have to steal as well as a win condition as well as relevant UI. I learned a lot about working with other's code, a deeper understanding of the navmesh, its parameters and how to implement it, and especially for my disguise, NPC AI. I also got to work on UI more which I seldomly do, I got to learn especially about implementing scripts for the UI.

Austin: What I did for the project was implement the player movement and the Guards as well as their state machine logic, detection zone, and patrol routes. Finally I did the captured screen when you are caught. Overall I learned a lot more about how to use the navmesh and the AI components of unity. I also learned how to set up and structure a statemachine to make the guard AI.